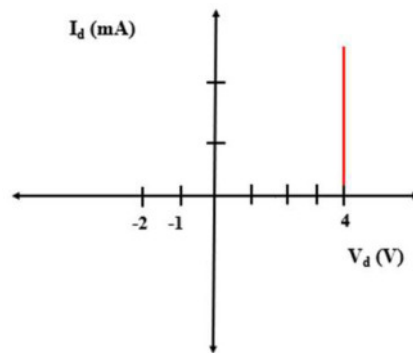
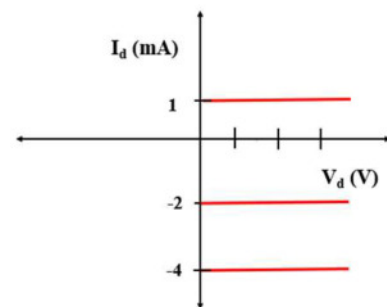
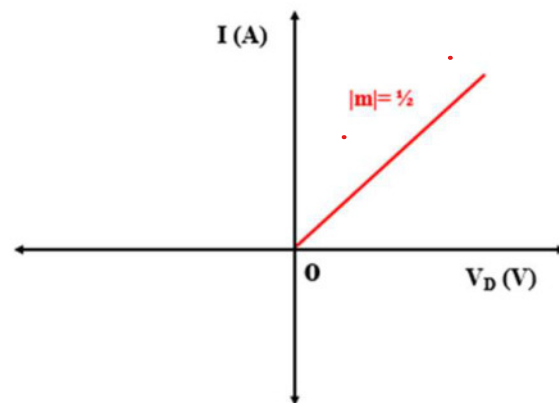
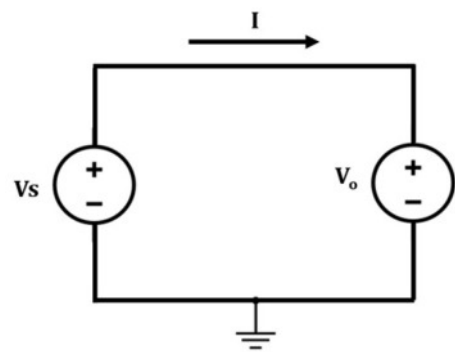
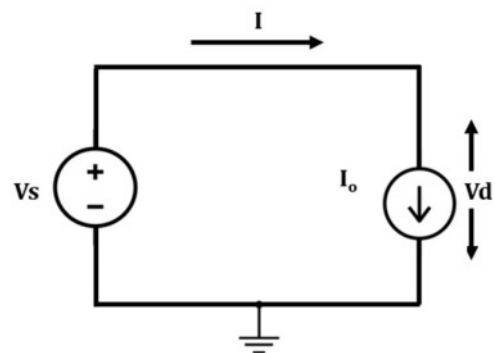
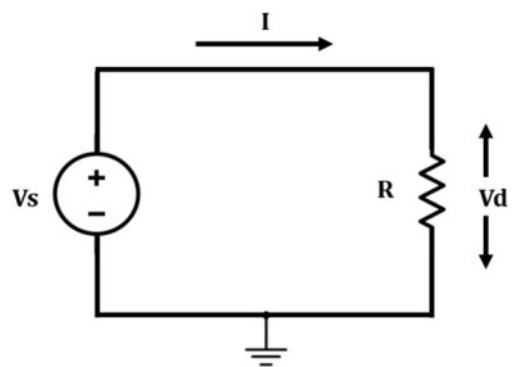


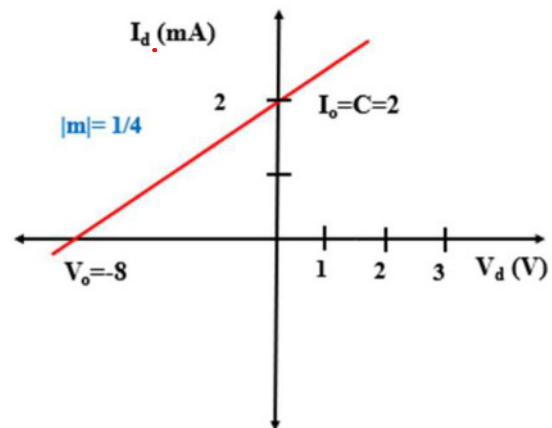
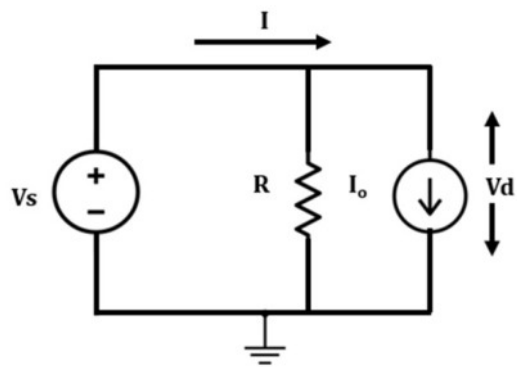
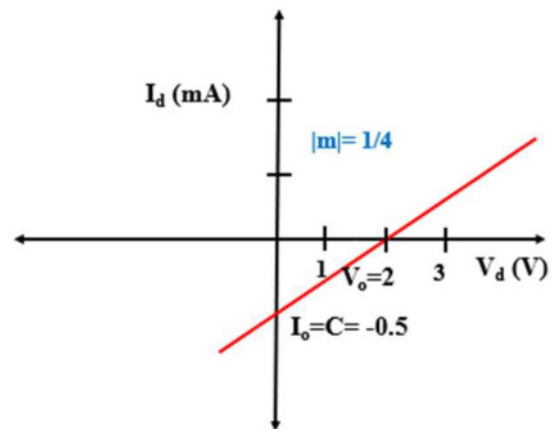
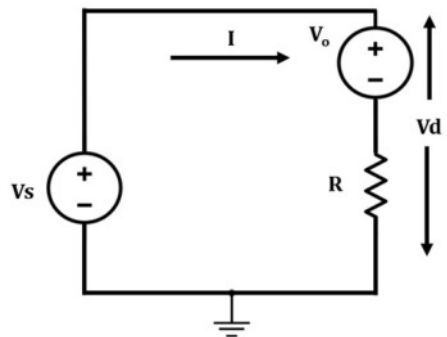
Introduction to Diode and Diode Logic Gates

Course No: CSE 251

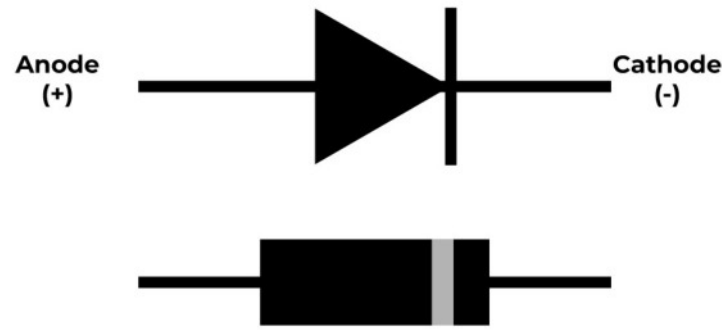
Course Title: Electronic Devices and Circuits

Prepared by: Aroni Ghosh (ARG)
Lecturer, CSE, Brac University



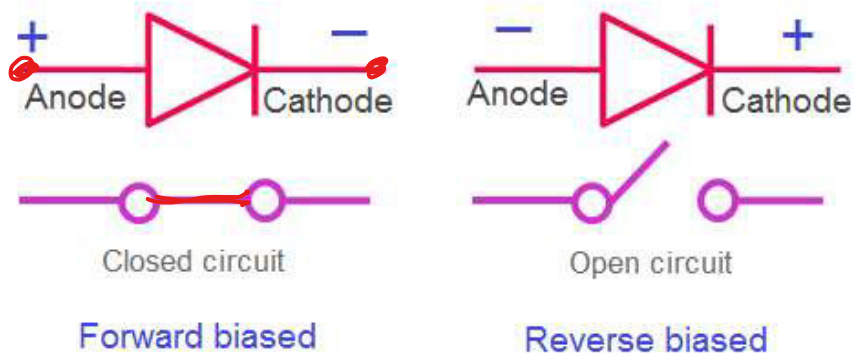


Introduction to Diode



- A two terminal Non-linear device. The two terminals are: Anode (+) and Cathode(-)
- The voltage difference between the terminals is expressed as, $V_d = V_{\text{anode}} - V_{\text{cathode}}$
- The working principle of a diode can be expressed with an “Electronic Valve”. It only allows the current to flow in one direction (From Anode to Cathode). This means the diode is “ON”/ in Forward Bias only when V_d is positive ($V_{\text{anode}} > V_{\text{cathode}}$). This property makes it suitable as a “Rectifier”

Ideal diode model and IV characteristics



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In the Ideal diode model, the following conditions are assumed:

- i. $V_{\text{anode}} > V_{\text{cathode}}$:
 - The Diode will be ON - It will act as a "Short circuit"
- ii. $V_{\text{anode}} < V_{\text{cathode}}$:
 - The Diode will be OFF
 - It will act as a "Open circuit"

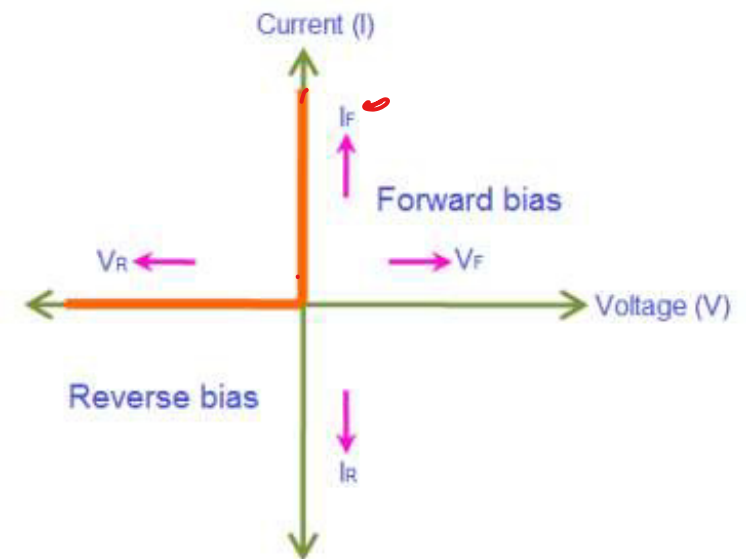


Fig: V-I characteristics of ideal diode

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Open circuit → I-V: x axis
Short Circuit → I-V: y axis

Real Diode I-V

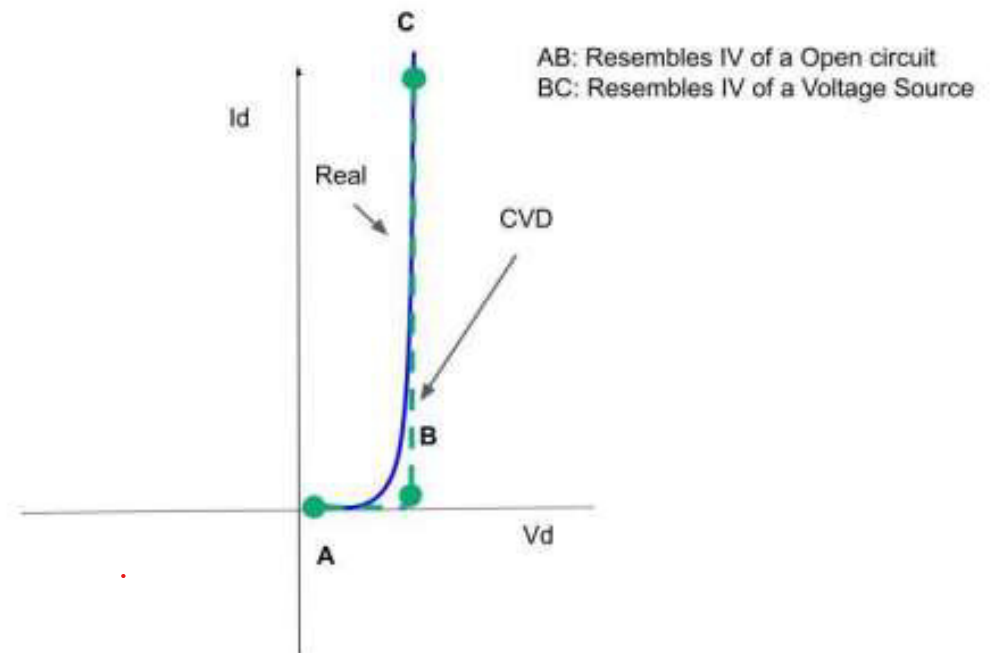
In the Real diode model, the following conditions are assumed:

i. $V_d > V_{\text{threshold}}$:

- The Diode will be ON
- It will act as a **“Constant Voltage Source”**

ii. $V_d < V_{\text{threshold}}$:

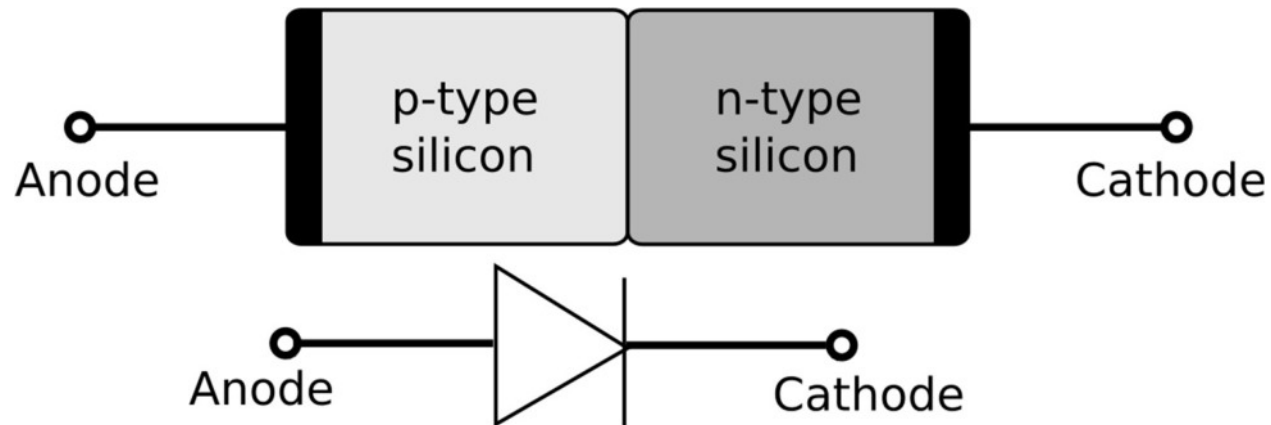
- The Diode will be OFF
- It will act as a **“Open circuit”**



Open circuit $\rightarrow$ I-V: x axis

Constant Voltage Source $\rightarrow$ I-V: parallel to y axis

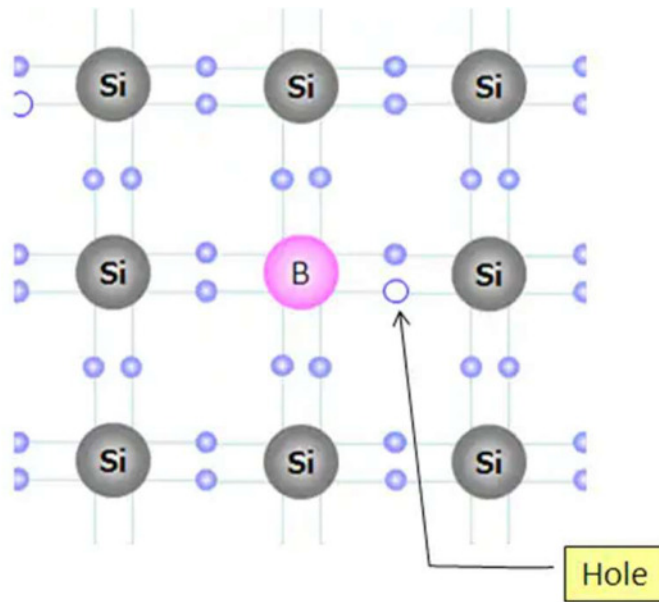
p-n junction Diode



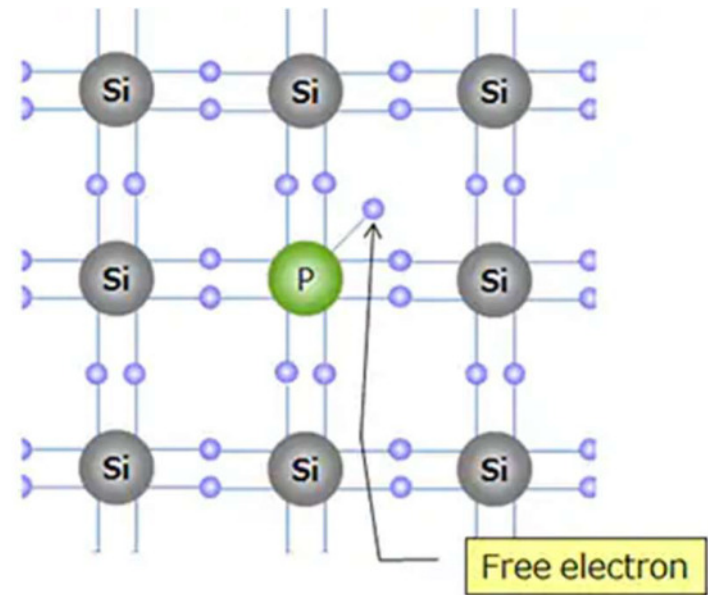
P-type semiconductor: When an intrinsic or pure semiconductor material (Like, Silicon (Si) and Germanium (Ge)) doped with acceptor impurities atom (Trivalent atoms, like Boron (B), Gallium (G), Aluminium(Al))

N-type semiconductor: When pure semiconductor doped with donor impurities (Pentavalent atoms, like Phosphorus (P), Arsenic (As), Antimony (Sb))

p, n type semiconductor

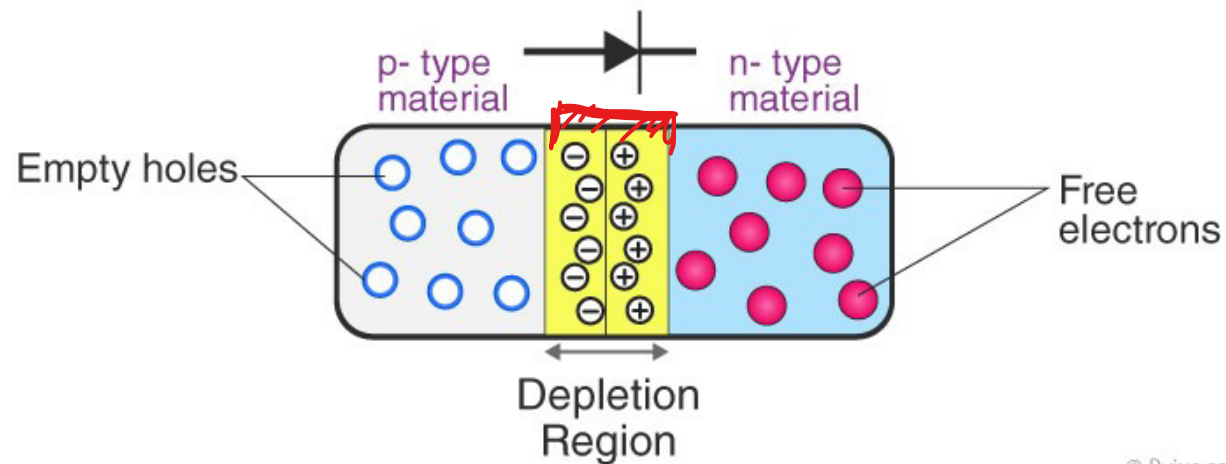


p- type



n- type

p-n junction



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Threshold voltage- for silicon about 0.7V
- for germanium about 0.3V.

Shockley Diode Equation

$$I_D = I_S \left(e^{\frac{qV_D}{nkT}} - 1 \right)$$

$$i_D = I_s \left[\exp\left(\frac{v_D}{nV_T}\right) - 1 \right] \quad V_T = \frac{kT}{q}$$

$k = 1.38 \times 10^{-23}$ J/K is Boltzmann's constant and $q = 1.60 \times 10^{-19}$ C is the magnitude of the electrical charge of an electron. At a temperature of 300 K, we have $V_T \cong 26$ mV

Where V_T is Thermal Voltage

2

Where I_S is reverse saturation current

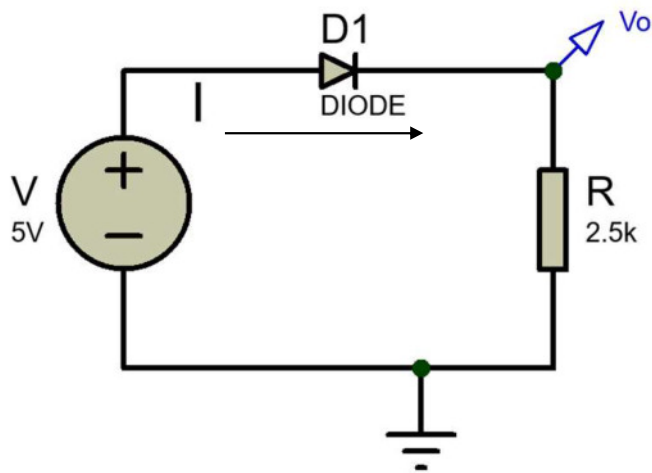
I_D and V_D are the diode current and voltage, respectively

q is the charge on the electron

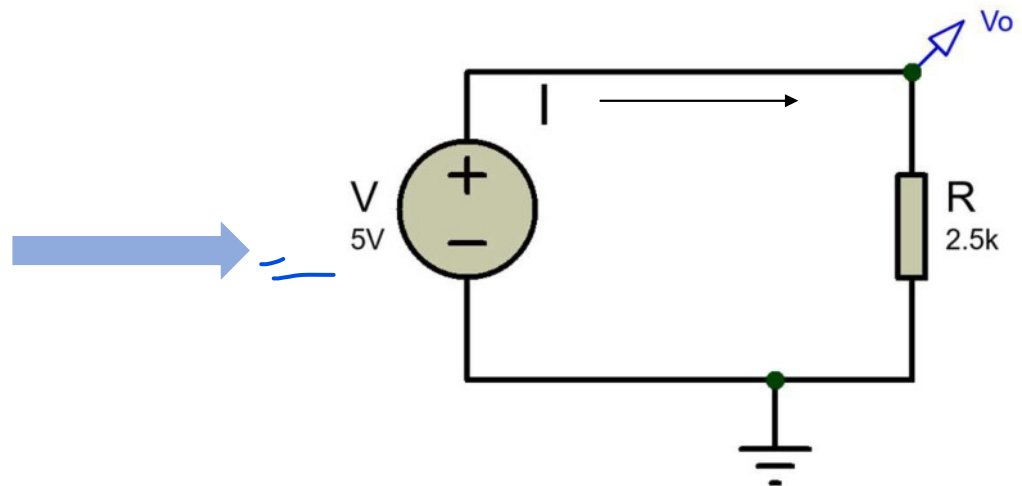
n is the ideality factor

Example Problems (Ideal Diode)

Example 1:



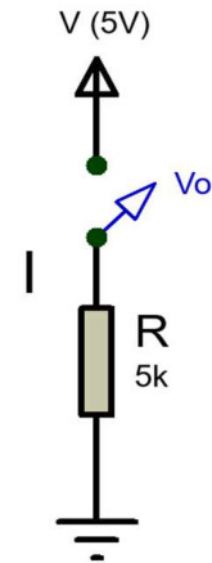
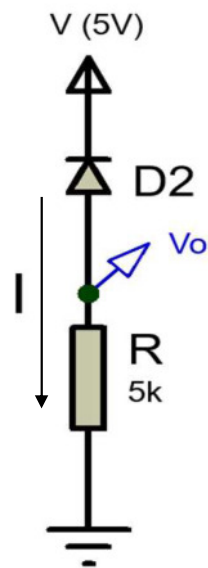
$V_{\text{anode}} = 5V$
Diode is on $\rightarrow$ replace with short circuit (**Ideal Diode Model**)



$V_o = V = 5V$
 $I = V_o / R = 5V / 2.5k\Omega = 2 \text{ mA}$

Example Problems (Ideal Diode)

Example 2:



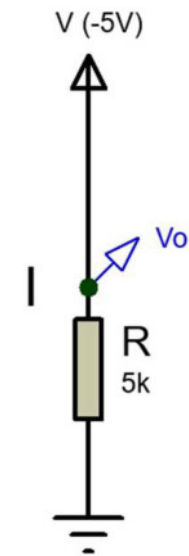
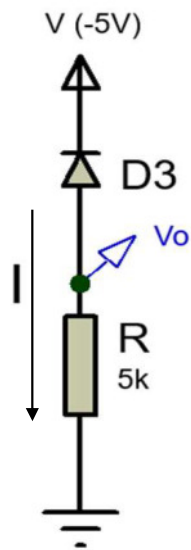
$$V_{\text{cathode}} = 5V$$

Diode is off $\rightarrow$ replace with open circuit (Ideal Diode Model)

As the circuit is open, $I = 0 \text{ mA}$
 $V_o = 0V$

Example Problems (Ideal Diode)

Example 3:



$$V_{\text{cathode}} = -5V$$

Diode is on $\rightarrow$ replace with short circuit (Ideal Diode Model)

$$V_o = -5V$$

$$I = V_o / R = -5V / 5k\Omega = -1 \text{ mA}$$

Logical Operations with Diode (OR)

Logic Truth Table

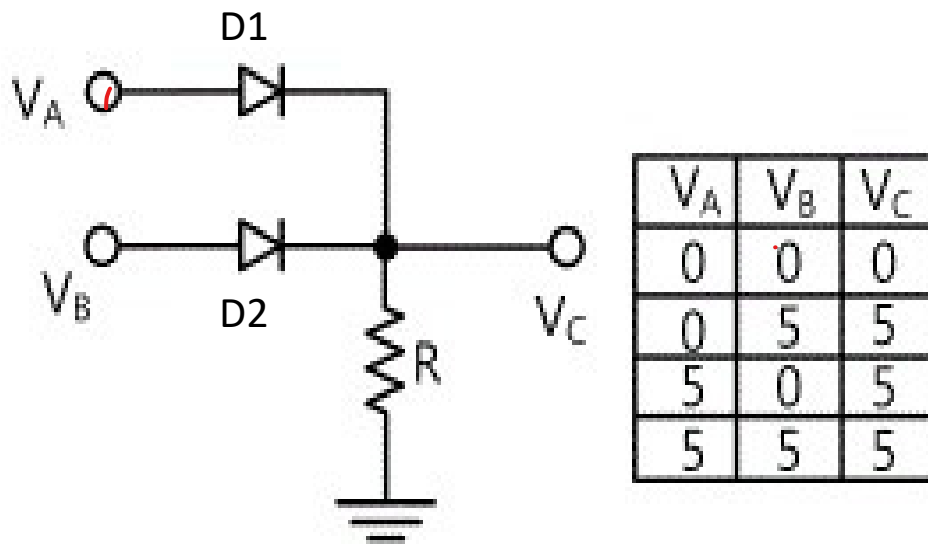
INPUTS		OUTPUT
X	Y	Z
0	0	0
0	1	1
1	0	1
1	1	1

Voltage Truth Table

INPUTS		OUTPUT
X	Y	Z
0 V	0 V	0 V
0 V	5 V	5 V
5 V	0 V	5 V
5 V	5 V	5 V

	Low/False	High/True
Logic Levels:	0	1
Corresponding voltage levels:	0V	5V

Logical Operations with Diode (OR)



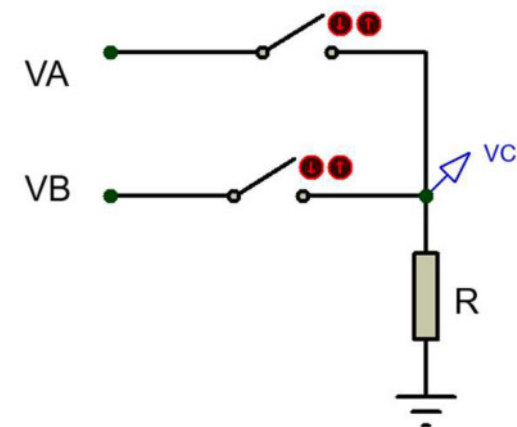
R is a **Pull Down Resistor** which pulls down the voltage of V_C from floating condition to 0V.

Case 1: $V_A = 0V$, $V_B = 0V$

Both D1 and D2 are off as $V_{\text{anode}} < V_{\text{cathode}}$

Replace D1 and D2 with open circuits.

So, $V_C = 0V$



Logical Operations with Diode (OR)

Case 2: $V_A = 0V$, $V_B = 5V$

D1 is off and D2 is on

Replace D1 with open circuit and D2 with short circuit.

So, $V_C = V_B = 5V$

Case 3: $V_A = 5V$, $V_B = 0V$

D1 is on and D2 is off

Replace D1 with short circuit and D2 with open circuit.

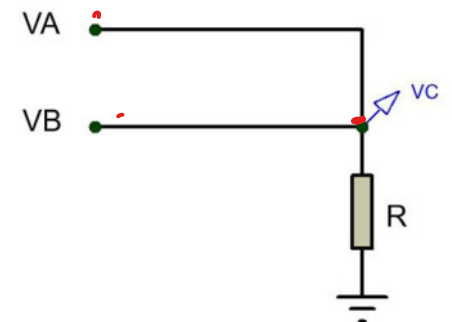
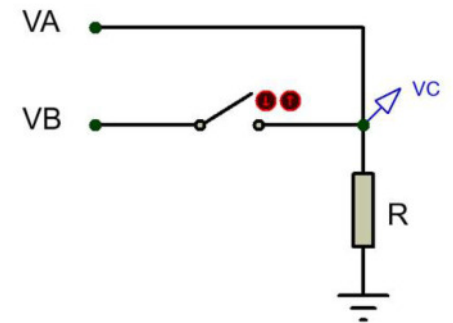
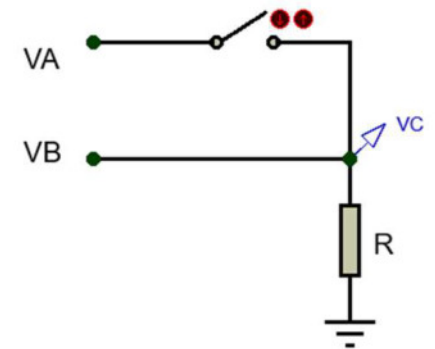
So, $V_C = V_A = 5V$

Case 1: $V_A = 5V$, $V_B = 5V$

Both D1 and D2 are on

Replace D1 and D2 with short circuits.

So, $V_C = V_A = V_B = 5V$



Logical Operations with Diode (AND)

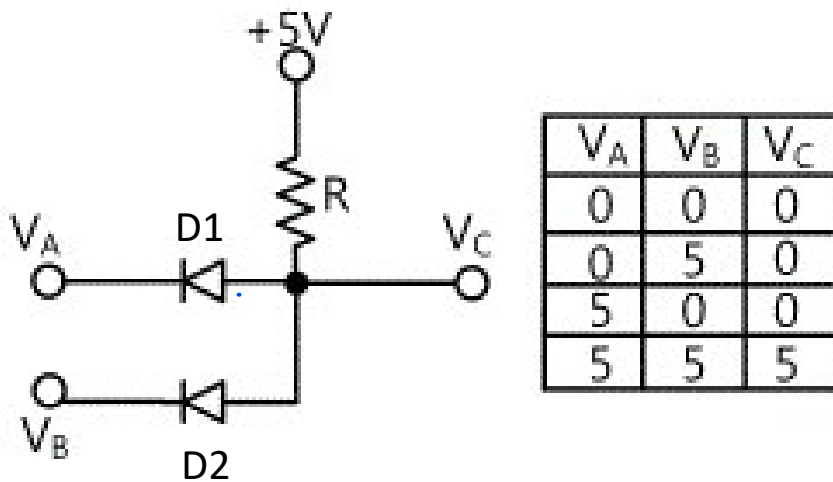
Logic Truth Table

INPUTS		OUTPUT
X	Y	Z
0	0	0
0	1	0
1	0	0
1	1	1

Voltage Truth Table

INPUTS		OUTPUT
X	Y	Z
0 V	0 V	0 V
0 V	5 V	0 V
5 V	0 V	0 V
5 V	5 V	5 V

Logical Operations with Diode (AND)



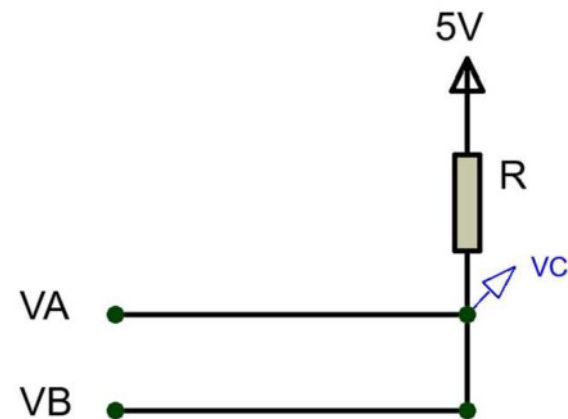
R is a **Pull Up Resistor** which pulls up the voltage of V_C from floating condition to 5V.

Case 1: $V_A = 0V$, $V_B = 0V$

Both D1 and D2 are on as $V_{\text{anode}} > V_{\text{cathode}}$

Replace D1 and D2 with short circuits.

So, , $V_C = V_A = V_B = 0V$



Logical Operations with Diode (AND)

Case 2: $V_A = 0V$, $V_B = 5V$

D1 is on and D2 is off

Replace D1 with short circuit and D2 with open circuit.

So, $V_C = V_A = 0V$

Case 3: $V_A = 5V$, $V_B = 0V$

D1 is off and D2 is on

Replace D1 with open circuit and D2 with short circuit.

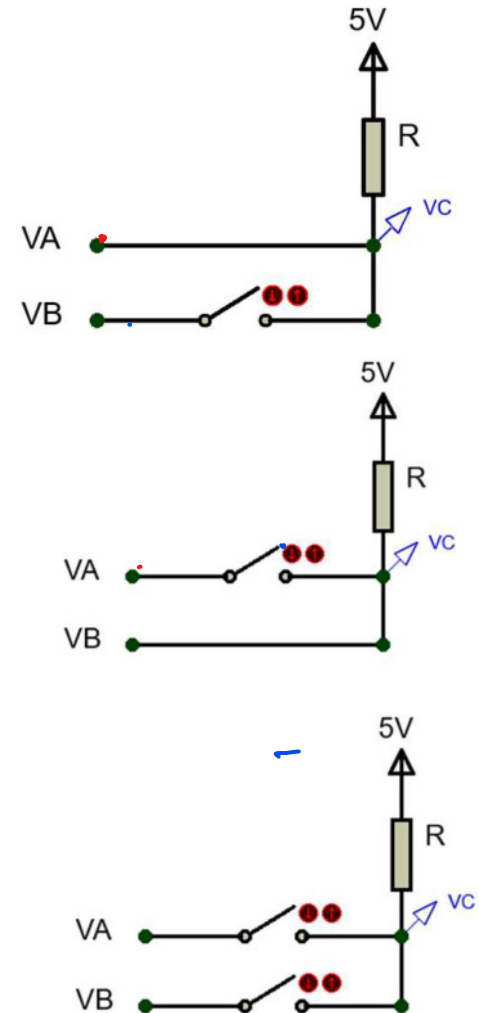
So, $V_C = V_B = 0V$

Case 1: $V_A = 5V$, $V_B = 5V$

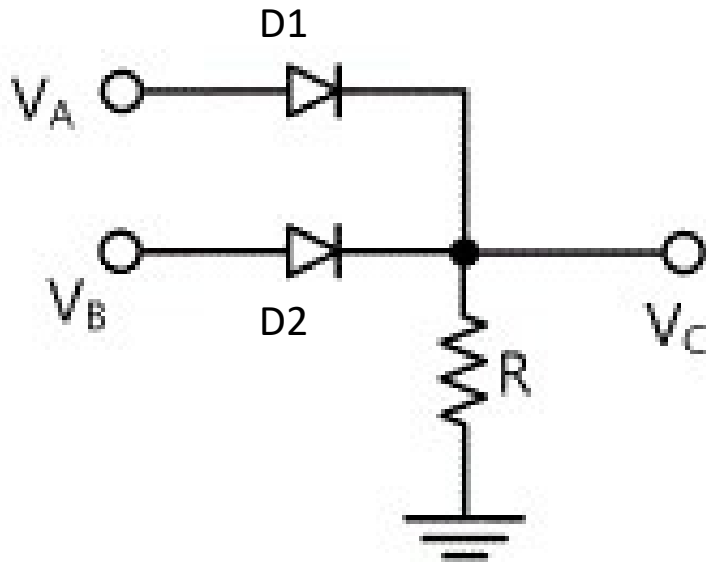
Both D1 and D2 are off

Replace D1 and D2 with open circuits.

So, $V_C = 5V$

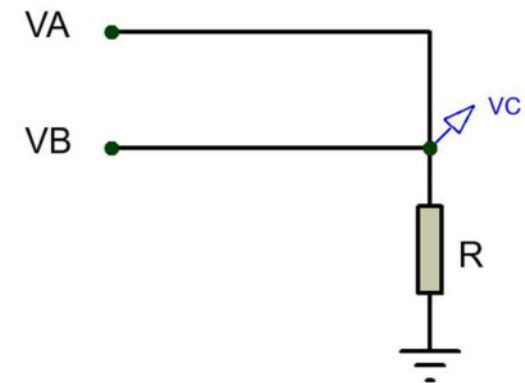


Effect of input Voltage Variation in Logic Gates (OR)



If $V_A = 4V$ and $V_B = 5V$, what is the value of output V_C ?

Case 1: Assuming both $D1$ and $D2$ are on
Replace $D1$ and $D2$ with short circuits.
So, V_C is short with both V_A and V_B
Can V_C be 4V and 5V at the same time???
So this assumption is **wrong!**



Effect of input Voltage Variation in Logic Gates (OR)

Case 2: Assuming both D1 and D2 are off

Replace D1 and D2 with open circuits. So, $V_C = 0V$

But now $V_{\text{anode}}(4V \text{ for D1, } 5V \text{ for D2}) > V_{\text{cathode}}(0V)$ for both diode

So this assumption is also **wrong!**

Case 3: Assuming D1 is on and D2 is off

Replace D1 with short circuit and D2 with open circuit. So, $V_C = V_A = 4V$

But now for D2, $V_{\text{anode}}(5V) > V_{\text{cathode}}(4V)$. So, D1 should have been **ON!**

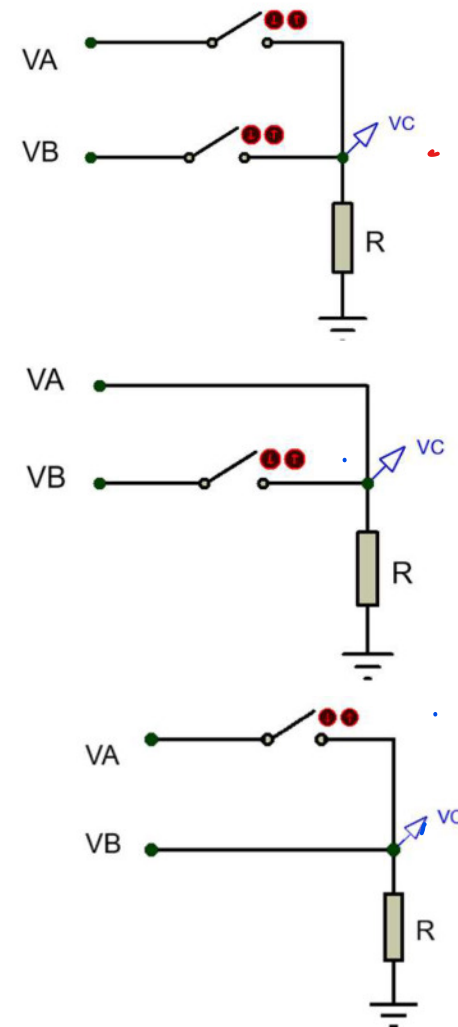
So this assumption is also **wrong!**

Case 4: Assuming D1 is off and D2 is on

Replace D1 with open circuit and D2 with short circuit. So, $V_C = V_B = 5V$

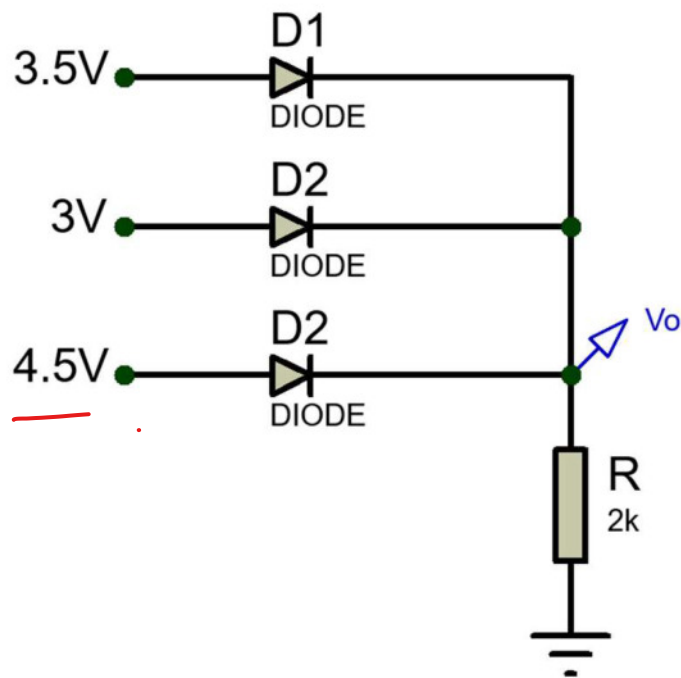
Now for D1, $V_{\text{anode}}(4V) < V_{\text{cathode}}(5V)$. So, D1 should be **OFF!**

So this assumption is **correct!**



Effect of input Voltage Variation in Logic Gates (OR)

Example 4: Find the value of V_o

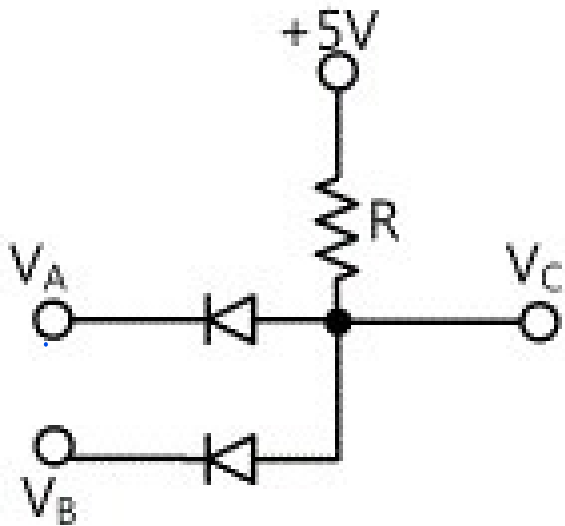


D1 and D2 are OFF $\rightarrow$ Open Circuit
Only D3 is ON $\rightarrow$ Short Circuit

$$V_o = 4.5V$$

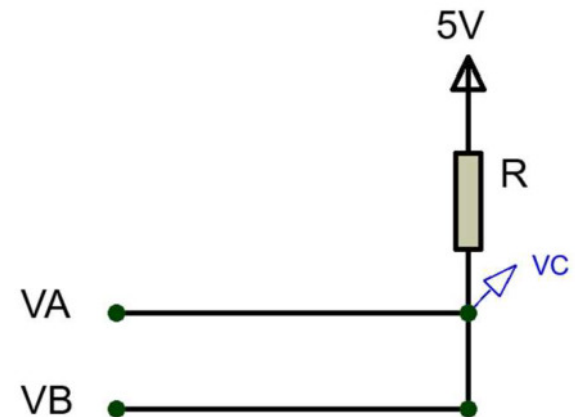
$$\text{Current, } I = 4.5V / 2k\Omega = 2.25 \text{ mA}$$

Effect of input Voltage Variation in Logic Gates (AND)



If $V_A = 1V$ and $V_B = 2V$, what is the value of output V_C ?

Case 1: Assuming both D1 and D2 are on
Replace D1 and D2 with short circuits.
So, V_C is short with both V_A and V_B
Can V_C be 1V and 2V at the same time???
So this assumption is **wrong!**



Effect of input Voltage Variation in Logic Gates (AND)

Case 2: Assuming both D1 and D2 are off

Replace D1 and D2 with open circuits. So, $V_C = 5V$

But now $V_{\text{anode}}(5V) > V_{\text{cathode}}(1V \text{ for D1 \& 2V for D2})$ for both diode

So this assumption is also **wrong!**

Case 3: Assuming D1 is off and D2 is on

Replace D1 with open circuit and D2 with short circuit. So, $V_C = V_B = 2V$

But now for D1, $V_{\text{anode}}(2V) > V_{\text{cathode}}(1V)$. So, D1 should have been **ON!**

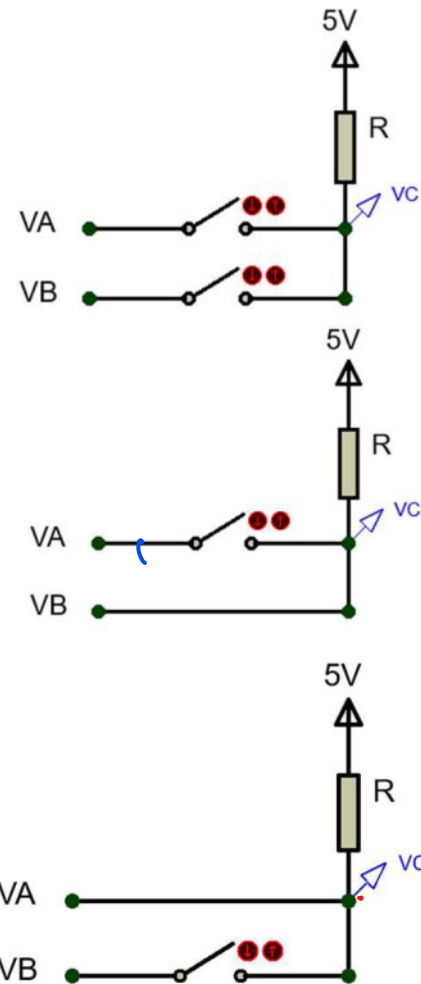
So this assumption is also **wrong!**

Case 4: Assuming D1 is on and D2 is off

Replace D1 with short circuit and D2 with open circuit. So, $V_C = V_A = 1V$

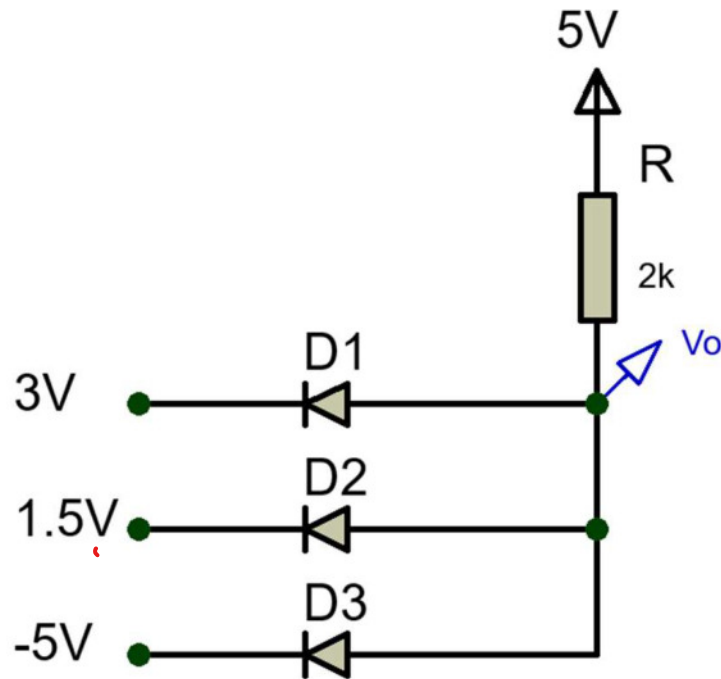
Now for D2, $V_{\text{anode}}(1V) < V_{\text{cathode}}(2V)$. So, D2 should be **OFF!**

So this assumption is **correct!**



Effect of input Voltage Variation in Logic Gates (AND)

Example 5: Find the value of V_o



D1 and D2 are OFF $\rightarrow$ Open Circuit
Only D3 is ON $\rightarrow$ Short Circuit

$V_o = -5V$

$$\text{Current, } I = \frac{5V - (-5V)}{2k\Omega} = 5 \text{ mA}$$

Thank You